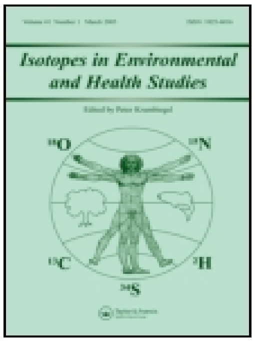

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Fertilizer nitrogen isotope signatures

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There has been considerable recent interest in the potential application of nitrogen isotope analysis in discriminating between organically and conventionally grown crops. A prerequisite of this approach is that there is a difference in the nitrogen isotope compositions of the fertilizers used in organic and conventional agriculture. We report new measurements of $\delta^{15}N$ values for synthetic nitrogen fertilizers and present a compilation of the new data with existing literature nitrogen isotope data. Nitrogen isotope values for fertilizers that may be permitted in organic cultivation systems are also reported (manures, composts, bloodmeal, bonemeal, hoof and horn, fishmeal and seaweed based fertilizers). The $\delta^{15}N$ values of the synthetic fertilizers in the compiled dataset fall within a narrow range close to 0‰ with 80% of samples lying between −2 and 2‰ and 98.5% of the data having $\delta^{15}N$ values of less than 4‰ (mean = 0.2‰ $n = 153$). The fertilizers that may be permitted in organic systems have a higher mean $\delta^{15}N$ value of 8.5‰ and exhibit a broader range in $\delta^{15}N$ values from 0.6 to 36.7‰ ($n = 83$). The possible application of the nitrogen isotope approach in discriminating between organically and conventionally grown crops is discussed in light of the fertilizer data presented here and with regard to other factors that are also important in determining crop nitrogen isotope values.

Keywords: Natural isotope variations; Nitrogen-15; Organic fertilizers; Synthetic fertilizers

1. Introduction

In 2004, the market value of organic products worldwide reached US$ 27.8 billions (€23.5 billions) with the largest share of organic products being sold in North America and Europe. The global market for organic produce and the amount of land managed organically worldwide are both expected to increase for the foreseeable future [1]. Although the rate of growth of the organic sector across the European Union has slowed since its rapid expansion during the 1990s, the North American market for organic products is reporting the highest rates of growth worldwide [1].

Stringent guidelines govern the inputs and practices to which organic farmers must adhere before their produce may be sold as 'organic'. Lampkin [2] discusses in detail the elements

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of organic farming methods and highlights the need for organic production to be viewed as more than a system of production that includes or excludes certain inputs. However, there is a widely held consumer perception that organic farming is farming without artificial fertilizers, pesticides and herbicides and few would dispute that an important attribute of organic production is that synthetic chemicals are not used.

One of the principles of organic farming is to minimize reliance on external inputs and to work as much as possible within a closed system with respect to organic matter and nutrient elements [2]. For many organic producers, particularly mixed livestock and arable farmers, this may be achievable through systems of crop rotation, manuring, composts and the use of green manures. However, many growers supplement the fertility of their soils through the use of additional fertilizers and soil conditioners.

The term 'organic' although often applied to fertilizers is, in this context, at best ambiguous, and at worst, open to misuse. Ambiguity has arisen because the word 'organic' can refer to products that are derived from living matter, or in the organic chemistry sense, to compounds which contain carbon (and hydrogen), as well as being used to describe a system of agriculture. There is no legal definition of 'organic' when applied to fertilizers and composts. A manufacturer could apply it to a product because it was produced by organic farming methods or simply because it was produced from something that once lived. The use of all fertilizers in organic systems is subject to the need for use being authorised by the appropriate inspection body or inspecting authority.

Fertilizers and soil conditioners that may be permitted in organic cultivation under Council Regulation (EEC) No 2092/91 are summarized in table 1. The use of many of these fertilizers and soil conditioners is often subject to additional conditions as specified in Regulation 2092/91 [3]. For example, the use of sawdust, wood chips, composted bark and wood ash are all subject to the condition that the wood being used has not been chemically treated after felling. The UK Compendium of Organic Standards [4] stipulate that in the first instance, the fertility and biological activity of soil must be maintained or enhanced by (a) cultivation of legumes, green manures or deep-rooting plants in an appropriate multi-annual rotation programme,

Table 1. Fertilizers and soil conditioners that may be permitted in organic cultivation in the European Union where a need is recognized by an appropriate certifying authority and subject to any additional conditions specified in Council Regulation (EEC) No 2092/91 [3].

- Farmyard manure
- Poultry manure
- Compost
- Slurry, urine etc.
- Peat (limited to horticulture)
- Clays (*e.g.* perlite, vermiculite)
- Mushroom culture wastes
- Dejecta of worms (vermicompost) and insects
- Guano
- Products or by-products of animal origin (*e.g.* blood meal, hoof meal, horn meal, bone meal or degelatinized bone meal, fish meal, meat meal, feather, hair and 'chiquette' meal, wool, fur, hair, dairy products)
- Products or by-products of plant origin, *e.g.* oilseed cake meal, cocoa husks, malt culms *etc.*
- Seaweeds and seaweed products
- Sawdust and wood chips
- Composted bark
- Wood ash
- Soft ground rock phosphate
- Aluminium calcium phosphate
- Basic slag
- Crude potassium salt (*e.g.* kainit, sylvinite)
- Potassium sulphate, possibly containing magnesium salt
- Stillage and stillage extract
- Calcium carbonate of natural origin (*e.g.* chalk, marl, ground limestone, Breton ameliorant (maërl), phosphate chalk)
- Magnesium and calcium carbonate of natural origin (*e.g.* magnesian chalk, ground magnesium limestone)
- Magnesium sulphate (*e.g.* kieserite)
- Calcium chloride solution
- Stone meal
- Calcium sulphate (gypsum)
- Industrial lime from sugar production
- Elemental sulphur
- Trace elements
- Sodium chloride (only mined salt)

(b) incorporation of livestock manure from organic livestock production, (c) incorporation of other organic material, composted or not, from holdings producing according to the rules of the UK Organic Standards. Other organic or mineral fertilizers (such as those listed in table 1) may, exceptionally, be applied if adequate nutrition of the crop is not possible by these methods.

The use of synthetic nitrogen fertilizers commonly used in conventional agriculture is prohibited in organic farming. It has been suggested that a 'fingerprint' of a synthetic nitrogen fertilizer may be left on crops grown using a synthetic fertilizer in the form of the stable nitrogen isotope composition of the crop. Several experiments in which crops have been cultivated under controlled conditions have demonstrated that crops grown using synthetic nitrogen fertilizers have lower nitrogen isotope values than crops grown using manure-based fertilizers [*e.g.* 5, 6]. Further, it has been suggested that this may provide a means for discriminating between organically and conventionally grown crops [7, 8].

The nitrogen isotope compositions of synthetic nitrogen fertilizers are well documented in the literature, [*e.g.* 9–14]. Kendall [15] comments that synthetic nitrogen fertilizers generally have nitrogen isotope values between −4 and 4 ‰. Kendall [15] reports that organic fertilizers including green manures, composts and liquid/solid animal waste, generally have higher δ^{15}N values and a much wider range of compositions (2–30 ‰) than synthetic fertilizers reflecting their more diverse origins. The nitrogen isotope composition of manures and composts are quite well established [*e.g.* 10, 12, 16]. The nitrogen isotope compositions of other types of fertilizer that may be permitted in organic systems are not well documented although some measurements are reported [*e.g.* 13, 17].

In this study, we report new measurements of nitrogen isotope values for various commercial synthetic fertilizers sourced in the UK and present a compilation of the new data with existing literature nitrogen isotope data. Nitrogen isotope values for fertilizers that may be permitted in organic systems (manures, composts, bloodmeal, bonemeal, hoof and horn, fishmeal and seaweed based fertilizers) are also presented.

2. Methods

Samples for nitrogen isotope analysis were freeze-dried and homogenized during grinding to a fine powder using a ball mill. Larger samples (*e.g.* manure, compost) were homogenized and sub-sampled before freeze-drying. Nitrogen isotope composition was determined using a PDZ Europa ANCA-GSL elemental analyser connected to a PDZ Europa 20:20 continuous flow isotope ratio mass spectrometer. Samples were analysed in triplicate and values accepted when precision (σ^{n-1}, $n = 3$) was <0.3 ‰. Dried samples were weighed into tin capsules, and standards and samples matched to give 0.1 or 0.2 mg (±20%) N per analysis. The N-content (% N on a dry weight basis) for the synthetic fertilizers was always known since this is specified by the manufacturer. Matching the total amount of nitrogen in samples and standards was therefore straightforward and 0.2 mg of N was analysed, an optimal amount for the IRMS used in this study. The N-content of the other fertilizers (manures, composts) was ascertained by elemental analysis before samples were prepared for mass spectrometric analysis. The N-content of some of these materials was often low (1–2%) compared with the much higher N-content of the synthetic fertilizers (usually 10–45%). For the low N-content materials, an amount of material sufficient to produce 0.1 mg of N was analysed. This reduced the amount of sample material per analysis, increased the chances of complete combustion of the samples, and helped to regulate the build up of ash on the combustion column. Total N-content of standards and samples were closely matched in order to minimize errors associated with source-linearity effects. Nitrogen isotope data are reported in conventional δ-notation in units of per mil (‰) with respect to atmospheric nitrogen (air) according to equation (1).

$$\delta^{15}\mathrm{N}_{\mathrm{sample}}(‰) = \left(\frac{R_{\mathrm{sample}} - R_{\mathrm{standard}}}{R_{\mathrm{standard}}}\right) \times 1000 \quad (1)$$

where, $R = {}^{15}\mathrm{N}/{}^{14}\mathrm{N}$ and the standard is atmospheric nitrogen with a $^{15}\mathrm{N}/^{14}\mathrm{N}$ ratio of 0.00368 and a δ^{15}N value of 0 ‰. Inorganic fertilizer samples were referenced against IAEA-N1 (ammonium sulphate reference material certified by the International Atomic Energy Agency) with a δ^{15}N composition of 0.4 ‰. Organic (carbon based) samples were referenced against a casein in-house standard with an accepted value of 6.3 ‰ (previously calibrated against IAEA reference materials during an inter-laboratory comparison exercise involving six European Food Control laboratories as part of European Union Project SMT4-CT98-2236 to develop methods to determine the origin of milk, butter and cheese). Long-term performance of the mass spectrometer was monitored by analysis of a secondary reference material in every batch. The secondary reference materials for inorganic and organic samples were IAEA-N2 (ammonium sulphate) and L-alanine (in-house reference) with accepted δ^{15}N values of 20.4 ‰ and 8.7 ‰, respectively. The long-term standard deviation of the values obtained from measurements of the secondary laboratory standards was 0.16 ‰.

3. Results and discussion

3.1 *Synthetic fertilizers*

The mean value for synthetic fertilizers analysed in this study was −0.2 ‰ ($\sigma^{n-1} = 2.1$ ‰) and except for a sample of ammonium sulphate with a δ^{15}N of 6.6 ‰, the synthetic fertilizer samples have δ^{15}N values <2.6 ‰, see tables 2 and 3. Nitrogen isotope values within a few per mil of zero reflect the source of nitrogen in these fertilizers, which is atmospheric nitrogen (with a δ^{15}N value of 0), and the efficiency of the fertilizer manufacturing process which results in very little difference in the nitrogen isotope composition of the starting material (atmospheric nitrogen) and the nitrogen in the fertilizer products.

The anomalous ammonium sulphate fertilizer with a δ^{15}N of 6.6 ‰ gave the same value (within error) on reanalysis. Ammonium sulphate is often produced by direct synthesis from ammonia produced by the Haber–Bosch process and sulphuric acid [18]. However, 60% of the global production of ammonium sulphate is as a by-product and co-product of other manufacturing processes, *e.g.* production of nylons and plastics, hydrometallurgy and treatment of flue gases [18]. The value of 6.6 ‰ obtained for the sample of ammonium sulphate may reflect the production of this sample as a by- or co-product of one of these other manufacturing processes where reaction conditions were such that the ammonium sulphate produced is isotopically heavier than that produced from direct synthesis by the Haber–Bosch process.

In our study, we did not measure the isotope composition of nitrate-N and ammonium-N separately in the products in which both species occur. Analysis of variance using the statistical package SPSS 14.0.0 showed that there was no significant difference between the mean values of the fertilizers containing nitrogen as ammonium-N, nitrate-N and urea-N ($p = 0.08$). Freyer and Aly [14] determined a mean difference between fertilizer nitrate-N and ammonium-N of 3.9 ‰ for samples that contained nitrogen in both the ammonium and nitrate form. Ammonium-N was isotopically lighter than the nitrate-N for all the samples analysed [14]. Shearer *et al.* [13] also found that ammonium-N was consistently isotopically lighter than nitrate-N in six ammonium nitrate samples obtained from different manufacturers. Vitoria, *et al.* [9] in a survey of

Table 2. Fertilizer composition and nitrogen isotope composition of synthetic nitrogen fertilizers.

	Fertilizer type	Manufacturer	$\delta^{15}N(‰)_{air}$
Synthetic	$(NH_4)_2SO_4$	W.L Dingley	0.8
	$(NH_4)_2SO_4$	Gem	6.6
	$(NH_4)_2SO_4$	Terra	−1.2
	$(NH_4)_2SO_4$	Bunn	0.7
	KNO_3	W.L Dingley	−1.5
	KNO_3	Gem	−1.1
	KNO_3	Yara	−1.0
	Urea	Gem	−2.4
	Urea	Unknown	−1.1
	Urea	Bunn	−0.8
	Urea	W.L Dingley	−1.6
	Urea	Yara	−5.9
	NH_4NO_3	Unknown	2.6
	NH_4NO_3	Bunn	0.5
	NH_4NO_3	Kemira	2.2
	NH_4NO_3	Terra	−1.3
	NH_4NO_3	Yara	−1.4
	NH_4NO_3	Unknown	−0.9
	$(NH_4)H_2PO_4$	Gem	−0.9
	$(NH_4)H_2PO_4$	Terra	−0.3
	NPK 20-10-10	Kemira	1.9
	NP 27-10	Kemira	0.8
	NPK 28-5-5	Kemira	1.1
	$Ca(NO_3)_2$	Yara	−0.3
	NPK 16-16-16	Yara	−0.6
	NPK 21-8-11	Yara	−0.7
	NPK 12-12-12	Unknown	0.4
	Hydroponic solution NH_4NO_3 and $Ca(NO_3)_2$	Unknown	0.2
	Hydroponic solution KNO_3 and $Ca(NO_3)_2$	Unknown	0.7

The fertilizer manufacturer is shown where known.

literature values report a median nitrate-N that is slightly enriched in ^{15}N (median value = 1.8 ‰) compared to atmospheric nitrogen while the median value for ammonium-N was slightly depleted in ^{15}N compared to atmospheric nitrogen (median value = −0.6 ‰). Freyer and Aly [14] suggest that ammonium-N is depleted with respect to ^{15}N compared to atmospheric nitrogen because the lighter ^{14}N isotope is kinetically favoured during the synthesis of ammonia. In the manufacture of nitrate fertilizers, ammonia from the Haber–Bosch process is vaporized, mixed with air and burned over a platinum/rhodium alloy catalyst. Nitrogen oxides (both NO and by-product N_2O) and water are formed and the NO is further oxidized to NO_2 before combining with water to give nitric acid. Nitric acid produced in this way is used to manufacture ammonium nitrate, calcium nitrate and potassium nitrate. Freyer and Aly [14] suggest that isotopic fractionation means that the waste gases from nitric acid production are depleted in ^{15}N and this leaves the residual nitric acid, from which the nitrate fertilizers are formed, isotopically slightly enriched in ^{15}N. Evidence for this is provided by Freyer and Aly from the nitrogen isotope composition of sodium nitrate produced from the waste nitrogen oxides of the nitric acid process ($\delta^{15}N = -22.7$ ‰).

3.2 *Fertilizers that may be permitted in organic agriculture*

Of the fertilizers analysed in this study, the farmyard manures have the highest mean $\delta^{15}N$ value (8.1 ‰) and largest range (3.5–16.2 ‰), see tables 3 and 4. Over time, animal manures tend to become isotopically heavier because volatilization of ^{14}N ammonia is kinetically favoured

Table 3. Summary of nitrogen isotope values $\delta^{15}N(‰)_{air}$ for fertilizers measured in this study.

	Mean	Standard deviation	Minimum	Maximum	*n*
Synthetic	−0.2	2.1	−5.9	6.6	29
Manure/compost	8.1	3.9	3.5	16.2	11
Seaweed based	2.5	1.5	0.6	5.4	9
Mammalian, non-manure	5.9	1.0	4.1	6.8	9
Fish based	7.1	3.6	2.1	10.6	4

Table 4. Nitrogen isotope composition of fertilizers that may be permitted in organic cultivation systems.

	Fertilizer type	Source	$\delta^{15}N(‰)_{air}$
Manure/composts	Farmyard manure	Grower	8.5
	Farmyard manure	Grower	9.3
	Farmyard manure	Grower	3.5
	Farmyard manure	Grower	7.2
	Farmyard manure	Grower	14.1
	Farmyard manure	Grower	16.2
	Farmyard manure + compost	Grower	4.9
	Farmyard manure	Grower	6.9
	Chicken manure pellets	Rooster	5.4
	Chicken manure pellets	Westland	4.8
	Chicken manure pellets	Grower	8.4
Seaweed based	Seaweed extract	Grower	1.5
	Seaweed extract	Maxicrop	1.3
	Seaweed based liquid feed	Greenfingers	2.9
	Seaweed based liquid feed	Vitax	3.1
	Natural seaweed meal	Maxicrop	5.4
	Seaweed based liquid feed, Type 1	B&Q	0.6
	Seaweed based liquid feed, Type 2	B&Q	1.7
	Seaweed based fertilizer, Type 1	Grower	3.7
	Seaweed based fertilizer, Type 2	Grower	2.1
Mammalian/non manure	Dried blood	W.L. Dingley	4.1
	Dried blood	Gem	6.8
	Dried blood	J. Arthur Bowers	6.6
	Dried blood	Vitax	6.3
	Hoof and horn	W.L.Dingley	6.2
	Hoof and horn	Gem	6.6
	Hoof and horn	Grower	6.3
	Bonemeal	Gem	5.1
	Bonemeal	J. Arthur Bowers	4.7
Fish based	Fish, blood and bone	J. Arthur Bowers	7.0
	Fishmeal	Grower	10.6
	Fishmeal-based	Nugro 8.7.7	2.1
	Fishmeal-based	Nugro 6.1.3	8.7

The fertilizer manufacturer is shown where known. Some fertilizers were supplied from 'Growers' and this is indicated on the table.

over volatilization of ^{15}N ammonia. It is possible that the manures with the lower $\delta^{15}N$ values were 'fresher' than those with the higher $\delta^{15}N$ values.

The manures noted as being from 'Growers' were obtained from certified organic growers and were manures authorized for application to their crops by their certification body. One of the pelleted chicken manure products is a UK Soil Association endorsed product. Its use by an organic grower must still be approved by an organic certification body but the Soil Association endorsement is an assurance that the poultry manure comes from farms that follow a high standard of humane animal husbandry. This does not apply to all poultry manure-based products, some of which may be derived from manure from intensively reared poultry. There are unlikely to be large differences in the δ^{15}N of manures from poultry produced under different production systems. The δ^{15}N of these manures will be governed by the source of dietary protein and this is unlikely to differ substantially between organic and conventional production systems, being dominated in both cases by vegetable protein from pulses, soya or peas/beans with a small amount of fishmeal used as a supplementary source of protein when necessary. For organically reared poultry, a high proportion and where possible, all of the feedstuffs must have been produced organically [19]. The feeding of mammalian meat and bonemeal to any farmed livestock (organic/conventional) including poultry is prohibited in the UK [20]. Some differences in the nitrogen isotope composition of poultry manures may result from the manure-ageing effects described above or during processing which usually includes sterilization and drying steps before milling into pellets.

The mammalian-based (non-manure) fertilizers cluster quite closely around a mean value of 5.9 ‰. The nitrogen isotope composition of animal protein materials including bone, hair and muscle protein are principally determined by diet and are generally enriched by 2 to 3 ‰ relative to the δ^{15}N composition of the dietary protein [21]. This means that the δ^{15}N of an animal-based (non-manure) fertilizer is likely to be determined mainly by the trophic level of the animal from which it derives. For example, fertilizer products from animals that are herbivores would be expected to be isotopically lighter than products made from animals that are carnivores. This could explain the relatively large range (from 2.1 to 10.6 ‰) in the fishmeal fertilizers if the different products are derived from fish that are at different trophic levels. Alternatively, it could be due to a difference in how the products are processed. The seaweed-based fertilizers have a mean value of 2.5 ‰ and range between 0.6 and 5.4 ‰. Seaweeds cannot fix nitrogen (N_2) and primarily take up N as nitrate and ammonium. Many seaweed-based fertilizer products are produced from the seaweed *Ascophyllum nodosum* which is harvested from inter-tidal waters of the North Atlantic and Arctic Oceans. One of the seaweed-based fertilizer producers states that their product is made from seaweed that is taken fresh from the sea and chopped, dried and milled into seaweed meal, a process that minimizes losses of volatile compounds that may otherwise be lost if the seaweed is left to dry naturally. Gaseous nitrogen-containing compounds lost during the drying process are likely to be isotopically relatively depleted in ^{15}N and such losses would result in an increase in the proportion of ^{15}N in the remaining fraction. Other processing may involve hydrolysing macerated seaweed in a pressure chamber to produce seaweed extract. Some manufacturers carry out this process under cold conditions and others use hot alkaline conditions. Variations in the δ^{15}N of seaweed-based fertilizer products may be due to variability in the processing of the seaweed or alternatively may result from variations in the δ^{15}N of the seaweed itself which in turn may reflect variable sources of the nitrate and ammonium taken up by the seaweed.

3.3 *Compilation of measured data with values from other studies*

Nitrogen isotope values for synthetic fertilizers and for fertilizers that may be permitted by EU Regulation 2092/91 are shown in figure 1 (data collated from refs. [8–14, 16, 17, 22–33]). Most of the synthetic nitrogen fertilizer samples fall within a relatively narrow range close to 0 ‰ with 80% of the samples lying between -2 ‰ and 2 ‰ and 98.5% of the data having δ^{15}N values of less than 4 ‰ (mean $= 0.2$ ‰, $\sigma^{n-1} = 1.9$ ‰, $n = 153$). The fertilizers that may be permitted in organic agriculture have a higher mean δ^{15}N value of 8.6 ‰ and exhibit a broader range in δ^{15}N values, from 0.6–36.7 ‰ ($n = 83$).

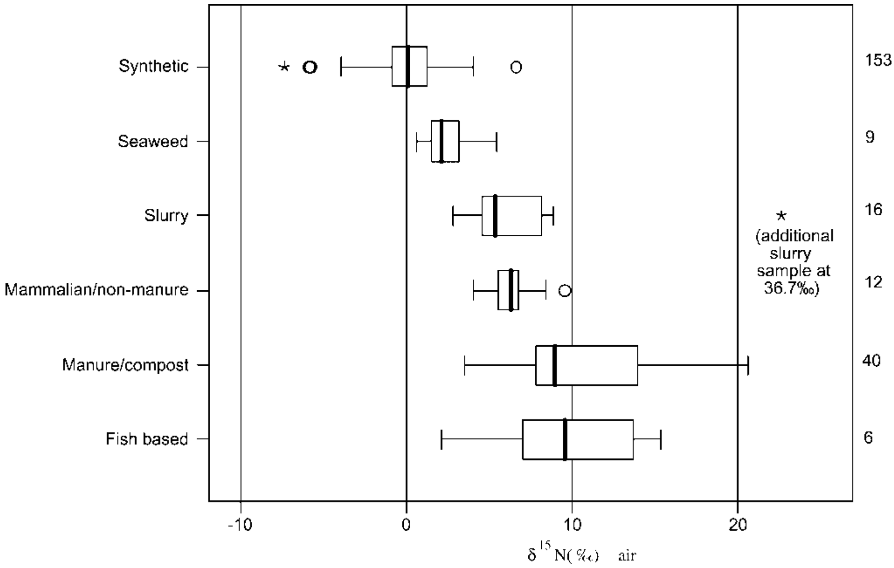

Figure 1. Box and whisker chart summarizing fertilizer data from this study and other published studies (see text for data sources). Boxes correspond to the interquartile range (IQ) containing the middle 50% of data. Whiskers indicate the highest and lowest values which are no greater than 1.5 times the IQ range. Bold lines across the boxes indicate median values. Outliers (o) are cases with values between 1.5 and 3 times the IQ range. Extremes (∗) are cases with values more than 3 times the IQ range. The number of samples '*n*' is indicated by the value at the right-hand side of the graph.

4. Further discussion

It is clear from the new measurements made in this study and published fertilizer $\delta^{15}N$ values that synthetic nitrogen fertilizers are typically more depleted in ^{15}N compared to the types of fertilizer that may be permitted in organic systems. In the compiled dataset, the mean $\delta^{15}N$ value for synthetic fertilizers was over 8 ‰ lower than the mean value for the types of fertilizer that may be permitted in organic agriculture. Although on average, synthetic nitrogen fertilizers tend to have lower $\delta^{15}N$ values there is a region of overlap with several of the fertilizers that may be permitted in organic systems having lower nitrogen isotope compositions than an ammonium sulphate fertilizer with a $\delta^{15}N$ value of 6.6 ‰.

The type of fertilizer used by a particular grower is likely to be strongly dependent on local soil conditions, the type of crop being grown, local agricultural practices and cost/availability of different types of fertilizer. For example, manures are more likely to be the fertilizer of choice for an organic farmer growing field-scale crops while applications of liquid fertilizers may be more common for high value horticultural crops. It is also important to emphasize that conventional growers may use synthetic fertilizers on their crops but they may also use any of the fertilizers permitted in organic farming.

As previously mentioned, it has been suggested that nitrogen isotope composition could be used to differentiate between organically and conventionally grown produce [7, 8]. The suitability of using the stable nitrogen isotope methodology in this way depends on the following factors.

(1) That the type and nitrogen isotope compositions of fertilizers generally used in organic and conventional agriculture are different.
(2) That the nitrogen isotope composition of the crop being cultivated is strongly influenced by the nitrogen isotope composition of the fertilizer applied.

Even if the first condition is generally met, there are many factors that are likely to influence the nitrogen isotope composition of the crop being grown. These include (a) the chemical form of the nitrogen in the fertilizer (*e.g.* NO_3^-, NH_4^+, organic-N), (b) the timing of the fertilizer application, (c) soil type and conditions, (d) other sources of nitrogen, *e.g.* atmospheric deposition, irrigation water, the use of leguminous crops (see [5] for a discussion), and (e) the cultivation history of the soil. For $\delta^{15}N$ to be a useful discriminator between organic and conventionally grown produce, the effect of fertilizer $\delta^{15}N$ on crop $\delta^{15}N$ must predominate over these other factors.

The $\delta^{15}N$ values for the fertilizers presented above are values for the total nitrogen in these samples. For the synthetic nitrogen fertilizers, the form of the nitrogen is known but for the fertilizers that may be permitted in organic systems, the fraction of the nitrogen present as organic N, NH_4^+, NO_3^- etc. is unknown. Even when the form of the applied nitrogen is known, isotopic fractionations associated with nitrogen transformations may occur before uptake by a plant, and the magnitude of these fractionations depends on the soil moisture and temperature conditions. Synthetic fertilizers are usually applied as 2–5 mm diameter prills/granules which are water soluble often releasing the nitrogen as the inorganic nitrogen species. After the prills have dissolved, any nitrate-N present in the prill is immediately plant-available. If the prill releases NH_4^+, then nitrification of ammonium-N to nitrate-N usually occurs before the nitrogen is taken up by plants. For prills made from synthetic urea, the nitrogen is present as the amide and under most soil conditions, the nitrogen is quickly hydrolized and the NH_4^+ readily converted (nitrified) to nitrate and quickly becomes plant-available.

Nitrogen transformations are accompanied by isotopic fractionations that can affect the nitrogen isotope compositions of the nitrogen species. For example, ammonia gas (NH_3) may form after the application of synthetic urea, and some losses of nitrogen to the atmosphere, particularly when soils are dry, may occur. When volatilization occurs, there is preferential loss of the light isotopologue of ammonia $^{14}NH_3$ compared to $^{15}NH_3$ leaving the residual nitrogen in the soil more enriched in ^{15}N. In contrast to synthetic fertilizers, manures and composts typically contain low amounts of soluble nitrogen and the organic nitrogen they contain undergoes mineralization before it is plant-available. Manures are also susceptible to losses of nitrogen through volatilization, and, as is the case for losses of ammonia after synthetic urea application, the volatilization will result in enrichment of ^{15}N of the nitrogen in the remaining fraction. Nitrogen cycling processes and their associated fractionations are one of the reasons why plant $\delta^{15}N$ values may not correspond with the nitrogen isotope value of an applied fertilizer. In addition, for soil-grown crops, plants will also use pre-existing nitrogen that is in the soil, other than that present from recent fertilizer application.

The suggestions by Nakano *et al.* [7] and Choi *et al.* [8] that nitrogen isotope composition of a crop might be a suitable discriminator between organic and conventionally grown produce were based on results of controlled cultivation experiments in which crops were grown using either synthetic nitrogen fertilizer or using a type of fertilizer that may be permitted in organic agriculture. In these experiments, other factors that may influence crop $\delta^{15}N$ were kept the same between treatments. Other studies have shown that timing of fertilizer application [29], soil moisture conditions [8], nitrate concentration and nitrogen isotope composition of nitrate in irrigation water [5] may all be influential in determining crop nitrogen isotope composition. Probably, the best assessment of whether the nitrogen isotope approach has any future as an appropriate tool to discriminate between organically and conventionally grown produce can be made from a survey of the $\delta^{15}N$ values of authentic commercially grown crops from across a wide geographical spread. Since there would be no reason to expect that any of the environmental factors would be systematically different between organic and conventional growers, any differences in the nitrogen isotope compositions of the organically and conventionally grown crops could therefore be attributed to the influence of fertilizer. The results of such a study are presented in Bateman *et al.* [34]. In this study, organically grown tomatoes and lettuces had mean nitrogen isotope values that were higher by 8.2 and 4.7 ‰, respectively, than their conventionally grown counterparts. No significant difference was found between the mean δ^{15}N values for organically and conventionally grown carrots. This is attributed to the cultivation of tomatoes and lettuces being relatively intensive, often under protected conditions, compared to field-grown carrots. This is consistent with findings by Schmidt *et al.* [17] who found significant differences between the δ^{15}N values of greenhouse-grown organic and conventional tomatoes and peppers but no significant difference between the δ^{15}N values of organically and conventionally cultivated wheat.

5. Conclusions

The nitrogen isotope values of synthetic fertilizers measured in this study and compiled with other literature values fall within a narrow range with a mean value of 0.2 ‰. Eighty percent of synthetic fertilizers had values between −2 and 2 ‰ with 98.5% of the fertilizers having δ^{15}N values of less than 4 ‰. The fertilizers that may be applied in organic systems had a higher mean δ^{15}N value of 8.6 ‰ and exhibited a much broader range in nitrogen isotope values from 0.6 to 36.7 ‰.

Although it is clear that there are differences in the nitrogen isotope composition of the fertilizers used in organic and conventional agriculture, and controlled cultivation experiments have shown that these differences impact the nitrogen isotope composition of a crop being grown, there are many other factors which can affect the nitrogen isotope composition of a crop. We contend that the best assessment of whether the nitrogen isotope approach is an appropriate tool to discriminate between organically and conventionally grown produce can be made from a survey of the δ^{15}N values of authentic commercially grown crops from across a wide geographical spread incorporating a broad range in environmental conditions and agricultural practices.

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